Quantum Cryptography and Post-Quantum Security

Submitted by:-

1. Anjani Kumar (240382)

Submitted to:-

Prof. Tabish Raseed

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Abstract

Over the last decade, rapid progress in quantum computing has raised serious concerns about the durability of the cryptographic systems that currently

safeguard digital communication, financial transactions, government data, and personal information. Many of today's widely used public-key algorithms—most notably RSA and Elliptic Curve Cryptography (ECC)—depend on mathematical problems that are difficult for classical computers but can be solved efficiently by sufficiently powerful quantum machines using algorithms such as Shor's and Grover's. Because of this emerging threat, researchers have focused on two broad strategies to ensure future-proof security. The first is Quantum Key Distribution (QKD), which uses fundamental quantum-mechanical properties to establish secret keys with information-theoretic security. The second is post-quantum cryptography (PQC), which develops new classical algorithms designed to withstand attacks from quantum computers.

This study examines and compares several influential QKD protocols, including BB84, E91, B92, and measurement-device-independent QKD, along with the major categories of PQC algorithms. The paper also reviews the underlying quantum concepts that make QKD possible, the historical shift from classical to quantum-resistant cryptographic models, and the practical risks posed by real-world quantum hacking strategies. A consolidated analysis outlines the trade-offs among performance, hardware dependencies, scalability, and security guarantees. The paper concludes by exploring ongoing challenges and future opportunities in the field—such as the development of quantum communication networks, satellite-assisted QKD, hybrid PQC-QKD security architectures, and the potential role of artificial intelligence in managing next-generation secure communication systems.

1. Introduction

In today's world, almost every aspect of daily life relies on some form of digital communication. Whether someone is checking their bank balance, placing an online order, sending an email, or accessing government services, their data is constantly being protected by cryptographic systems running in the background. Modern security on the internet largely depends on classical cryptography—systems such as RSA, Diffie–Hellman, and Elliptic Curve Cryptography (ECC) for secure key exchange and digital signatures, along with symmetric algorithms like AES for encrypting data. These techniques have worked effectively for decades because the mathematical problems they rely on, such as factoring large numbers or solving discrete logarithms, are extremely difficult for classical computers to solve within any practical timeframe.

This assumption, however, becomes invalid when considering the capabilities of future quantum computers. Quantum algorithms pose a direct threat to many of the cryptographic tools used today. For instance, Shor's algorithm can solve the integer factorization and discrete logarithm problems exponentially faster than the best known classical approaches, making RSA and ECC vulnerable. Grover's algorithm, although less destructive, still reduces the security of symmetric systems by speeding up brute-force attacks. As a result, once scalable quantum computers are developed, many widely adopted security mechanisms could become ineffective.

In response to this potential crisis, researchers around the world are exploring two major pathways to secure communication in the post-quantum era. One approach is Quantum Key Distribution (QKD), which makes use of the principles of quantum mechanics to create secret keys that cannot be intercepted without detection. The other approach—post-quantum cryptography (PQC)—focuses on designing new classical algorithms that are believed to resist attacks even from quantum computers. This paper examines both directions, comparing their strengths, limitations, and practical considerations. The goal is to provide a clear and accessible understanding of how these techniques contribute to building quantum-resilient security systems, especially for students beginning their study of quantum cryptography.

2. Background and Methods

2.1 Evolution of Classical Cryptography and Quantum Threats

Traditional cryptography has mainly developed along two broad categories: **symmetric-key** and **asymmetric-key** systems. In symmetric cryptography, a single secret key is shared between the sender and the receiver, and this same key is used for both encryption and decryption. Well-known examples include AES and, historically, DES. These systems are efficient and fast, but they depend heavily on a secure way to distribute the shared secret key.

Asymmetric or public-key cryptography solved this problem by introducing a public key for encryption and a private key for decryption. RSA and ECC are now deeply embedded in almost every secure communication platform—HTTPS websites, VPNs, messaging apps, and many authentication systems. Their security relies on mathematical problems that appear extremely hard for classical computers. For example, RSA depends on the difficulty of factorizing a large composite number, something that becomes exponentially harder as the number grows. ECC works on an even more compact structure, relying on the challenge of solving discrete logarithms on elliptic curves.

The arrival of quantum computing, however, completely changes this security landscape. Quantum computers operate using an entirely different computational model, and certain mathematical problems that are infeasible for classical computers can be solved quickly with quantum algorithms. Shor's algorithm is the most notable example: it can factor large integers and compute discrete logarithms in polynomial time, effectively breaking RSA, ECC, and similar schemes. Grover's algorithm also weakens symmetric cryptography, though its impact is less severe; doubling key sizes often restores security. Because public-key algorithms break fundamentally under quantum attacks, researchers are now urgently developing new approaches that remain secure in a quantum-capable world.

2.2 Basic Quantum Mechanics for Cryptography

To understand how quantum cryptography works, a few core quantum concepts must be introduced, though they can be described intuitively. Unlike a classical bit, which is always either 0 or 1, a quantum bit—commonly called a *qubit*—can exist in a **superposition** of both 0 and 1. When a measurement is made on a qubit, this superposition collapses to a definite value according to specific probability rules.

A key property that underlies quantum cryptography is the **no-cloning theorem**, which states that unknown quantum states cannot be copied perfectly. This is very different from classical information, which can be duplicated without restriction. Because of this limitation, an eavesdropper cannot quietly intercept and copy quantum states used in a key distribution process.

Another important phenomenon is **entanglement**, where two particles become correlated in such a way that measuring one instantly affects the state of the other, even when they are separated by large distances. Protocols like E91 take advantage of these entangled states to enhance security. When combined—superposition, measurement disturbance, no-cloning, and entanglement—these quantum principles make it possible to detect any attempt to intercept or tamper with the communication channel.

2.3 Quantum Key Distribution Protocols

Quantum Key Distribution (QKD) does not replace classical encryption; instead, it provides a secure way to generate and share secret keys that can then be used with classical symmetric algorithms such as AES. Over the years, researchers have proposed and tested several QKD protocols, each with different mechanisms and advantages.

2.3.1 BB84 Protocol

BB84, introduced by Bennett and Brassard in 1984, is considered the foundational QKD protocol. In BB84, Alice uses single photons to encode bit values using two different bases—typically rectilinear (+) and diagonal (×) polarizations. She randomly chooses both a bit value and a basis when preparing

each photon. Bob, without knowing her choices, randomly selects bases to measure the incoming photons.

After the transmission, both parties publicly discuss which bases they used, discarding all cases where their bases differed. The remaining bits should match if no eavesdropping occurred. If an attacker, often referred to as Eve, attempts to intercept the photons, she introduces detectable errors because she must guess the measurement basis. A sufficiently high error rate signals the presence of eavesdropping, while a low error rate allows Alice and Bob to apply error correction and privacy amplification to produce a secure final key.

2.3.2 E91 Protocol

The E91 protocol, proposed by Ekert in 1991, relies on **quantum entanglement** rather than prepared single-photon states. A source generates entangled photon pairs and sends one photon from each pair to Alice and the other to Bob. They independently choose measurement settings and later use part of the data to test Bell's inequality. If Bell's inequality is violated, it confirms the presence of genuine quantum entanglement and ensures that no classical or hidden-variable explanation can account for their correlated results.

The remaining measurement outcomes—those not used for the Bell test—form the raw data from which a shared secret key can be extracted. While E91 is conceptually elegant and grounded in fundamental physics, practical implementation requires high-quality entanglement sources and stable transmission channels, which makes it more technically demanding.

2.3.3 B92 Protocol

The B92 protocol simplifies the QKD process by using only two non-orthogonal quantum states. Alice encodes bits using these states, and Bob performs measurements that produce either a conclusive result (where he can determine what Alice sent) or an inconclusive one. Because the two states cannot be

perfectly distinguished, an eavesdropper cannot reliably determine the transmitted bits without introducing disturbances.

Although B92 is simpler than BB84, its performance can be sensitive to noise and implementation choices. Under certain conditions, BB84 may offer better efficiency or robustness, but B92 remains an important theoretical alternative and is useful in scenarios where hardware constraints limit the available quantum states.

2.3.4 Measurement-Device-Independent QKD (MDI-QKD)

One of the biggest vulnerabilities in practical QKD systems lies in the detectors, particularly on the receiver's end. MDI-QKD eliminates this major weakness. In this approach, both Alice and Bob send quantum states to a third party—often called the measurement node or relay—which performs a joint measurement. Importantly, this relay does not need to be trusted; it might even be controlled by an adversary.

After the relay announces the measurement results, Alice and Bob can use that information, along with their knowledge of the states they sent, to derive a shared secret key. Because the protocol does not rely on trusting any measurement device, it provides strong security even against sophisticated detector-based attacks. This makes MDI-QKD a promising option for constructing secure, large-scale quantum networks.

2.4 Post-Quantum Cryptography

Post-quantum cryptography (PQC) aims to create classical algorithms that can withstand attacks from both classical and quantum computers. Unlike QKD, PQC does not require specialized quantum equipment, making it far easier to deploy across existing internet infrastructure.

Several families of PQC algorithms are being actively researched:

- **Lattice-based cryptography**, which relies on hard problems such as Learning With Errors (LWE). These algorithms show strong potential for both encryption and signatures, though they usually involve larger keys.

- **Code-based cryptography**, which uses problems from coding theory. The McEliece cryptosystem is a classic example known for extremely large public keys but very strong security.

- **Hash-based signatures**, which use cryptographic hash functions in novel ways. They offer strong security proofs but sometimes produce large signatures.

- **Multivariate and isogeny-based schemes**, which rely on polynomial equations or elliptic-curve isogenies. These fields are still evolving, and some schemes have recently been broken, showing how dynamic PQC research is.

Organizations like NIST are currently evaluating and standardizing PQC algorithms, an effort that is essential for guiding global adoption and ensuring long-term security.

2.5 Quantum Attacks on Classical Cryptography

The most direct quantum threat to modern cryptography comes from Shor's algorithm. Once quantum computers reach sufficient scale and stability, Shor's algorithm will be able to break RSA, Diffie–Hellman, and ECC by solving their underlying mathematical problems efficiently. This jeopardizes all encrypted data that relies on these schemes, including data already stored today that might be decrypted in the future once quantum computers mature.

Grover's algorithm poses a different but still significant risk. It speeds up brute-force search by reducing the effective security level of symmetric keys. A symmetric key that normally provides n bits of security effectively drops to $n/2$ bits under a quantum attacker. Increasing symmetric key sizes—for example, using AES-256 instead of AES-128—can effectively address this risk.

Although quantum computers cannot yet break widely used cryptographic systems, the “harvest now, decrypt later” threat has made the transition to quantum-safe cryptography urgent, especially for sensitive, long-lived data.

2.6 State-of-the-Art Quantum Communication Systems

QKD has progressed far beyond theoretical proposals. Numerous field trials have demonstrated successful quantum key exchange over existing fiber-optic networks spanning tens—or even hundreds—of kilometers. Under ideal conditions, QKD systems can achieve key rates ranging from a few kilobits per second to several megabits per second for short distances.

Free-space QKD has also advanced rapidly. Satellite-based experiments, including those using China’s Micius satellite, have demonstrated secure quantum communication over thousands of kilometers. These experiments show that global quantum communication is feasible, although they involve complex tracking systems and must account for atmospheric interference.

To extend QKD to even longer distances, researchers are developing **quantum repeaters** and **trusted-node networks**. While quantum repeaters remain experimental, trusted-node infrastructures have already been deployed in several metropolitan testbeds.

2.7 Methods Used in This Study

This research is grounded in a structured review of academic literature, including foundational papers, survey articles, and experimental reports on QKD, PQC, and quantum security challenges. Sources were selected from peer-reviewed journals, conference proceedings, and technical reports. Key protocols such as BB84, E91, and MDI-QKD were examined alongside the major families of post-quantum cryptographic schemes.

The study compares classical, quantum, and post-quantum security models based on several factors, including performance, underlying mathematical difficulty, implementation requirements, and practical suitability. Experimental results from existing QKD deployments were reviewed to understand real-world behavior, though the paper does not include independent simulations or

laboratory work. To clarify concepts, qualitative diagrams and simplified numerical examples were incorporated where helpful.

3. Results and Discussion

3.1 Classical vs Quantum-Secure Protocols

Classical public-key cryptography, post-quantum cryptography, and quantum key distribution each provide security in different ways and serve different practical needs. Algorithms like RSA and ECC have been used for decades because they are efficient, easy to integrate with current internet protocols, and well supported by existing hardware. They work smoothly with systems such as TLS for website security and VPN technologies for private communication. Their main weakness, however, is that their mathematical foundations collapse in the presence of powerful quantum computers capable of running Shor's algorithm.

Post-quantum cryptography (PQC) is intended to address this problem by introducing mathematical problems for which no fast quantum algorithms are known. Lattice-based, code-based, and other PQC families can be deployed using the same hardware and software infrastructure we use today. Even though these algorithms can require significantly larger keys or signatures, and may need updates to widespread protocols, they offer a viable path forward as quantum computers continue to evolve.

Quantum Key Distribution (QKD), in contrast, does not rely on computational difficulty at all. Instead, it offers **information-theoretic security**, meaning that its security holds even if an attacker has infinite computational power. This strength, however, comes with practical challenges: QKD requires specialized quantum transmitters and detectors, precise timing, and low-loss communication channels. Because of these constraints, deploying QKD at a global scale is currently expensive and technically demanding.

A realistic approach for the future may combine QKD and PQC. QKD can be used for highly sensitive communication links—such as inter-government networks or data-center backbones—while PQC algorithms can provide quantum-resistant security for the broader internet.

3.2 Performance Analysis of QKD Protocols

QKD performance is often evaluated through two metrics: the maximum distance over which it can operate securely and the key generation rate. Among the various protocols, BB84 remains one of the most widely implemented. It usually performs well over short to moderate distances, offering high key rates when optical loss is manageable. However, the longer the fiber link, the more the signal weakens, and at some point the system cannot reliably generate secure keys unless trusted repeaters or other intermediate infrastructure are used.

Entanglement-based protocols such as E91 generally provide stronger theoretical security but are harder to build in practice. Producing stable entangled photons and keeping them intact over long distances involves complex hardware and precise alignment. Similarly, the B92 protocol is conceptually elegant but often more sensitive to noise, making it less robust in real-world environments unless carefully tuned.

MDI-QKD typically produces lower key rates than BB84 because it requires more complex measurement operations and two simultaneous quantum channels. Despite this, its ability to remove vulnerabilities associated with measurement devices makes it a compelling choice for high-security networks. More recent innovations, such as twin-field QKD, aim to extend secure communication distances well beyond previous limits, offering key rates that grow more slowly with distance and helping to close the gap between theory and global-scale implementation.

3.3 Security Proofs and Practical Security

The mathematical foundations of QKD are extremely strong. Its security proofs describe how the quantum states behave, how an eavesdropper could theoretically interfere, and how classical post-processing steps reduce the adversary’s knowledge. If the quantum bit error rate stays below a protocol-specific threshold, the amount of information available to an attacker can be made exponentially small.

In practice, however, real systems are far from perfect. Hardware such as single-photon detectors may exhibit differences in detector efficiency, accidental

dark counts, or slight timing inconsistencies. Sources may fail to produce ideal single-photon states. These practical issues can create loopholes that do not appear in the theoretical models.

Modern QKD research therefore focuses strongly on addressing these imperfections. MDI-QKD eliminates a major category of vulnerabilities in measurement devices. Device-independent QKD goes even further by requiring almost no assumptions about the internal workings of the devices, though its experimental requirements are extremely demanding. As technology improves, these advanced security models may become more practical..

3.4 Experimental Implementations of QKD

Over the past twenty years, QKD has transitioned from a laboratory curiosity into a technology demonstrated in real urban environments. Many experiments have taken place over existing telecommunications fiber, showing stable secure key generation over tens of kilometers. Using enhanced detectors and decoy-state BB84, some systems have reached distances of over 100 km while still maintaining viability.

Larger-scale QKD testbeds connect several cities through a combination of fiber and trusted intermediate nodes. In these networks, keys are generated segment by segment and relayed securely through trusted stations. Although this requires confidence in the integrity of the intermediate nodes, it allows for long-distance QKD with existing technology.

Satellite-based QKD represents the most ambitious progress in the field. China's Micius satellite has demonstrated entanglement distribution, satellite-to-ground quantum key exchange, and even intercontinental secure communication. These experiments show that QKD can, in principle, operate at global distances, although the equipment needed for such missions remains costly and difficult to deploy.

3.5 Real-World Deployment Examples

The Micius satellite project is often cited as the strongest real-world demonstration of large-scale quantum communication. Its experiments, which

range from sending entangled photons across more than a thousand kilometers to enabling secure communication between distant countries, have proven that quantum communication can function reliably even outside controlled laboratory settings.

The DARPA Quantum Network was another significant milestone. It was one of the earliest attempts to link multiple QKD systems in an operational environment. The project explored challenges like key management, routing, and coexistence between quantum and classical communication channels, and laid the groundwork for later metropolitan-scale networks.

Commercial companies have also entered the field. Toshiba, among others, has developed QKD systems intended for enterprise use. These products aim for continuous operation, compatibility with existing IT infrastructure, and integration into standard encryption hardware. Such systems are already being evaluated for use in critical infrastructure, government communication, and financial networks.

3.6 Challenges: Noise, Decoherence, Losses and Hardware

Although QKD provides exceptional theoretical security, implementing it on a large scale remains difficult. Quantum signals are extremely fragile. In optical fiber, photons may be absorbed or scattered, and the loss increases as the distance grows. In free-space communication, factors such as weather conditions, turbulence, and precise alignment affect reliability.

Noise and decoherence also degrade the signal before it reaches the receiver. Single-photon detectors may fail to detect every photon and may even produce false detections due to dark counts. Small timing inaccuracies can introduce additional errors. As a result, practical systems must carefully adjust parameters such as pulse intensity, detector thresholds, and filtering to maintain acceptable performance.

The cost and complexity of the required hardware present additional barriers. Many detectors require cooling to very low temperatures, and sources of entangled photons are expensive to build and maintain. These factors make QKD more suitable for high-value communication channels rather than everyday use by the general public.

3.7 Quantum Hacking Techniques

Even though QKD protocols are theoretically secure, practical implementations can be vulnerable to side-channel attacks. Two of the most significant categories are **photon-number-splitting (PNS) attacks** and **detector-blinding attacks**.

PNS attacks exploit the fact that real QKD systems often use weak laser pulses that occasionally contain more than one photon. A well-equipped eavesdropper could capture one photon and let the others continue to the receiver, potentially gaining information without causing detectable errors. Decoy-state methods address this weakness by varying the intensity of pulses, making it difficult for the attacker to identify which pulses contain multiple photons.

Detector-blinding attacks manipulate the receiver's detectors using bright light, pushing them into a classical regime where the attacker can influence the detection outcomes. This allows an attacker to bypass the quantum behavior the protocol depends on. Experiments have shown that even commercial QKD systems can be vulnerable if not protected carefully.

Other attacks, such as Trojan-horse attacks (which send unwanted light into the system to gather internal information) and timing attacks, show that security requires more than correct mathematical proofs—it requires robust engineering and constant vigilance..

3.8 Countermeasures and Secure Designs

To address these vulnerabilities, researchers have developed a range of countermeasures. **Decoy-state QKD** is now widely used in systems that rely on weak coherent pulses. By analyzing the detection statistics of decoy and signal states, legitimate users can estimate how many detections came from single-photon events and identify anomalies that may indicate a PNS attack.

MDI-QKD provides an even stronger approach by rendering all detector-related vulnerabilities irrelevant. Since the measurement devices are treated as untrusted, attacks on the detectors cannot compromise the final key. Even if an adversary completely controls the measurement node, the key shared between the legitimate parties remains secure under the protocol's assumptions.

On the hardware side, improved detector designs, shielding, and real-time monitoring circuits help detect suspicious conditions such as attempted blinding or unexpected signal levels. Standardization bodies are also working to define testing procedures and certification standards so that QKD devices meet rigorous security requirements. These engineering and protocol-level improvements together move QKD closer to achieving the strong, ideal security promised by quantum theory.

4. Conclusions and Future Scope

4.1 Summary of Findings

This study examined quantum cryptography and quantum security from the broader perspective of building communication systems that can remain secure even in a future where quantum computers are widespread. The paper began by highlighting the weaknesses of today's public-key cryptographic methods, particularly RSA and ECC, when faced with quantum algorithms like Shor's and Grover's. These vulnerabilities create a strong motivation for exploring security technologies that cannot be undermined by quantum computation.

To address this, the paper introduced the basic quantum principles that make Quantum Key Distribution possible and reviewed several important QKD protocols, including BB84, E91, B92, and MDI-QKD. Alongside QKD, the study also discussed post-quantum cryptography as a classical but quantum-resistant alternative that relies on new mathematical foundations. A comparison of classical cryptography, PQC, and QKD showed that each approach offers its own balance of security, performance, hardware complexity, and ease of deployment.

Real-world demonstrations—such as long-distance fiber-based QKD links, satellite-based experiments, and early QKD networks like the DARPA Quantum Network—show that quantum-secure communication is no longer just theoretical. At the same time, actual implementation introduces challenges, and various quantum hacking demonstrations remind us that perfect theoretical security does not automatically translate into practical security. Techniques like decoy-state QKD, Device-Independent QKD, and secure hardware design are vital for closing the gap between theory and real-world practice..

4.2 Limitations of the Present Work

The findings presented in this paper are based largely on a survey of existing literature rather than hands-on experimental testing. The work does not include independent simulations or detailed mathematical derivations of security proofs, as the focus is on conceptual understanding rather than technical depth. Many

performance estimates and security arguments rely on previously published studies, which may differ depending on experimental conditions or assumptions.

Another important limitation is that quantum cryptography and post-quantum cryptography are evolving fields. New attacks, more efficient protocols, and updated recommendations are continually emerging. Because of this rapid development, some conclusions drawn here may become outdated as new research appears. A more technical or experimental study would be needed in the future to validate the concepts discussed in this paper under realistic conditions.

4.3 Future Scope

In the coming years, the vision of a **quantum internet**—a global network that allows quantum devices to exchange information securely—will continue to grow. QKD will likely be one of the foundational building blocks of such networks, enabling secure key exchange for both classical and quantum applications. However, reaching this level of functionality will require significant advancements in quantum repeaters, error-corrected quantum memories, and the reliable distribution of entanglement over long distances.

We may also see the rise of hybrid QKD networks that combine terrestrial fiber links, free-space optical communication, and satellite-based systems. These networks will need seamless integration with existing routing protocols, management software, and key distribution systems. Developing common standards and easy-to-use interfaces will be critical so that organizations can adopt quantum-secure technologies without needing in-depth knowledge of quantum physics.

Post-quantum cryptography is expected to advance alongside QKD. Once the NIST PQC standards are fully finalized and adopted, organizations will need clear migration roadmaps to replace vulnerable cryptographic algorithms in their infrastructure. Special attention will be necessary for data that must remain secure for many years, such as medical information, government records, and sensitive industrial data.

Finally, artificial intelligence and machine learning are likely to play a central role in managing and securing quantum-enabled networks. AI can monitor QKD channels for anomalies, optimize configuration and routing, and enhance threat detection in systems that combine QKD, PQC, and traditional cryptography. Together, these developments are expected to shape the next generation of secure communication, where classical and quantum technologies coexist and reinforce each other.

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